

HEMANTH RAJ M V

Bengaluru, Karnataka

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Education

BMS College of Engineering, Bengaluru
Bachelor of Engineering in Computer Science (Data Science)

Oct 2023 – Present
CGPA: 8.32

Technical Skills

Languages: Java, Python, SQL, Kotlin, C

ML & Data: Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, NLP (NLTK), EDA

Big Data & MLOps: PySpark, Apache Spark, Spark MLlib, FastAPI, REST APIs, Docker, RAG Pipelines

Databases & Tools: ChromaDB, PostgreSQL, MongoDB, Power BI, Tableau, Git, Jupyter Notebook

Projects

- **RiskGuard — Automated Return-Risk Decisioning Platform** *Razorpay AI Buildathon 2026*
FastAPI, Next.js/React, Tailwind CSS, LightGBM, SQLite, SHAP
 - Built a full-stack **internal decision-support tool** — **FastAPI** backend + **Next.js/React** frontend — that scores e-commerce return risk in real time, **automating a manual fraud-review process**.
 - Designed a **chronological 65/15/20 train/validation/test split** to eliminate threshold-selection leakage, and reconciled **SHAP** explainability output against **calibrated probability scores** for trustworthy real-time decisions.
 - Shipped a **9-item front-end/back-end iteration** — SSE-based live order feed, model-card analytics tiles, decisions-log dashboard — taking the tool from prototype to a **production-ready internal application**.
- **Tether — Social Accountability Android App** *GitHub — APK (v1.0)*
Kotlin, MVVM, Firebase Auth/Firestore/FCM, Navigation Component
 - Built a production-grade Android app using **Kotlin/MVVM**; optimized navigation with manual **NavController** handling to eliminate UI flickering and achieve **sub-second tab switching**.
 - Engineered a **Foreground Service** for multi-mode focus tracking (Stopwatch/Pomodoro) with **1-second polling accuracy**, ensuring **100% session persistence** under battery-restricted environments.
 - Implemented a **real-time nudge system** using **FCM + Firestore** with **sub-second sync**; optimized backend via **composite indexing** to efficiently handle thousands of log entries by group and date.
- **Visual Document Q&A — End-to-End RAG Pipeline** *GitHub*
FastAPI, ChromaDB, Ollama (phi3), PyMuPDF
 - Built a **Visual Document Q&A system** with a **FastAPI + ChromaDB** RAG pipeline using **all-MiniLM-L6-v2** embeddings and **Ollama (phi3)**, achieving **~12 tok/s** streaming via SSE with **1s retrieval latency**.
 - Engineered a **4-signal Concept Importance Engine** (TF-IDF + spread + heading weight + **PageRank centrality**) and visualized concept relationships as an **Obsidian-style D3.js force-directed graph** with tiered node coloring, co-occurrence edges, and click-to-query interaction.
 - Implemented **tiered heading detection** (H1/H2/H3 via PyMuPDF font-size analysis) and **async LLM topic refinement** that renders the TF-IDF graph instantly then merges domain concepts in background — boosting validated terms by **1.5×** with silent fallback ensuring zero UX disruption.

Achievements / Leadership

Nutanix Certified Associate (NCA) 2026

- Validated core competency in **hyperconverged infrastructure (HCI)** and **cloud infrastructure fundamentals**.

GenAI Hackathon — Special Mention 2025

- Recognized for **Visual Document Q&A**, a RAG-based system with FastAPI, ChromaDB, and a D3.js visualization layer.

Treasurer, DSynC Club — BMSCE April 2025 – Present

- Managed end-to-end club finances across **10+ events**, owning budgeting, expense tracking, and reimbursements. Secured corporate sponsorships for **UTSAV** (60K+ footfall) and **PHASESHIFT** (40K+ footfall)